Namrata patel

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AI/ML Engineer with 1 year of hands-on experience building production-grade deep learning systems for medical imaging and industrial automation. Skilled in object detection, segmentation, and classification using CNNs and Vision Transformers, with focus on healthcare AI.

Key Skills

Programming & Core Tech: Python, SQL, OpenCV, Git/GitHub.
DL & Computer Vision: Image Segmentation (U-Net), Vision Transformers (ViT), Object Detection (YOLOv8/v10)
Data Visualization: Excel, Google Sheets, SQL, Data Cleaning, Matplotlib, Pandas, Seaborn
Tools & Platforms:: Jupyter Notebook, VS Code, Google Colab
Soft Skills: Communication, Teamwork, Problem Solving, Presentation, Project Planning
Machine Learning Frameworks: PyTorch, Keras, Scikit-learn, TensorFlow

Professional Experience

Pixonate Lab Pvt. Ltd. — **Pune**
AI/ML Engineer (Full-Time) — Jun 2025 - Present

- Research & Development: Leading research-based projects utilizing Vision Transformers (ViT) for high-accuracy medical analysis and non-invasive diagnostics.
- Advanced CV Deployment: Implementing YOLO (v8/v10) for real-time object detection and integrating vision systems into industrial automation workflows.
- Healthcare AI: Developing complex classification and segmentation models for medical imaging, focusing on diagnostic reliability and model interpretability.

Projects

Non-Invasive Anemia Detection Using Vision Transformers - **Healthcare – Medical Imaging** [Project Link]
[U-Net, ViT-B/16, Streamlit, Data Cleaning, Cross-Validation, Automated Image Cleaning, Artifact Removal] — Dec 2025 - Jan 2026

- Engineered a three-stage pipeline using YOLOv8, U-Net, and ViT-B/16 for non-invasive diagnostic classification with 85.2% accuracy. Implemented date-wise sequential cross-validation and attention maps to ensure a leakage-safe, clinically interpretable medical AI system.

AI-Powered Tumor Localization in Whole Slide Images - **Healthcare – Computational Pathology / Medical Imaging**
[Python, PyTorch, ResNet18, UNet, CLAM, K-Means, Computer Vision, WSI Analysis] — Oct 2025 - Nov 2025

- Developed a computational pathology pipeline using ResNet18 and CLAM for attention-based tumor localization in high-resolution WSI datasets. Achieved 95%+ accuracy by optimizing embedding extraction with UNet, significantly enhancing diagnostic throughput and pathological focus.

Employee Monitoring System Using Face Recognition - **Computer Vision**
[Python, OpenCV, PyTorch, RetinaFace, YOLOv8, FaceNet, Streamlit] — Aug 2025 - Sep 2025

- Architected a real-time CCTV integration system leveraging YOLOv8 and FaceNet to automate attendance tracking with 95% identification accuracy. Streamlined operational reliability by deploying a Streamlit dashboard for high-precision, automated employee monitoring and reporting.

Education

Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere — **Pune**
Bachelor of Technology - Computer Science — Nov 2022 - Aug 2025
CGPA - 7.7

Maharashtra State Board of Technical Engineering — **Shahada**
Diploma - Computer Science — Nov 2019 - Jun 2022
CGPA - 8.3

ML/DL Pest Detection in Precision Agriculture [Link]

- **Published in the Indian Journal of Agriculture Engineering (IJAE)**, Engineered a multi-stage CNN pipeline using 2D convolution and max-pooling for accurate pest detection. The system enables early identification to reduce crop losses and support sustainable farming, validated using Accuracy, Precision, Recall, and F1-score.